

Clinical Summary for the Doctor

Principal genetic findings, level of evidence, pharmacogenetics and exact list of analyzes, exams, vaccines and screenings to be discussed

Sérgio F · 03-10-2026 · local time

Abstract for clinical use, based only on Sérgio F genes (genotyping chip, local imputation and polygenic scores). It is not a diagnosis or prescription: all actions are marked as a suggestion (not applied) and the decision is made by the doctor.

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1. Main message

Patient: Sérgio F. Source: consumption genotyping chip (strand +, GRCh38), local imputation (DR2 $\geq$ 0.8) and polygenic scores of PGS Catalog; percentiles in the reference population. It does not include blood tests or exams.

1. **Only plausible pathogenic type finding: SERPINA1 Pi*Z heterozygous** (rs28929474 C/T; Probable Pi*MZ). It needs confirmation with serum alpha-1-antitrypsin and Pi phenotype.
2. **Liver:** Pi*MZ + **PNPLA3 I148M** (rs738409 G/C) + **MBOAT7** rs641738 C/T; attenuated by **HSD17B13** rs72613567 T/TA (protector).
3. **Cardiovascular:** Homozygous 9p21 in the risk haplotype of high activity 5 SNPs and CETP, but low global coronary score (p5). Aortic stenosis score p89-95 and genetics of hypertrophic cardiomyopathy p76-89. Venous thrombosis p86.3 (loci p95.1; F11 T/T; F5 Leiden and prothrombin negative).
4. **High scores to be monitored:** glaucoma p97,9, emphysema p96,2, tinnitus p99,8, colorectal p94.3 (polyps p92.7; early onset p82.3), bladder cancer p94,9, p84 fractures, fragility p75 with lean mass p10, depression p95, ADHD p94/p97, insomnia p89, sarcoidosis p92.3.
5. **Actionable Pharmacogenetics:** CYP2C9 *1/*3, VKORC1 -1639 T/T, NAT2 slow, UGT1A1 *28, CYP3A5 expresser CYP2D6 without copy number; HLA-B*58:01 by typing (see the Pharmacogenetic Card).

All polygenic scores indicate relative risk, non-diagnosis.

2. Main findings and level of evidence

Area	Finding	Genotype / score	Evidence	Clinical implications
Alpha-1-antitrypsin	Probable Pi*MZ	rs28929474 C/T; rs17580 T/T (without S)	High (chip + concordant marker); lack of confirmation	Cofactor of liver fibrosis and emphysema with smoking or pollution
Fatty liver and fibrosis	PNPLA3 I148M; MBOAT7	rs738409 G/C; rs641738 C/T	High	Additive risk with Pi*MZ; HSD17B13 T/TA attenuates
Coronary disease	9p21 homozygous (5 SNPs in LD); CETP high activity	rs1333049, rs10757278 and others; rs708272 G/G	High (locus); global p5 score	ApoB/LDL as low as possible; no smoking
Lp(a)	High genetics	LPA rs3798220 T/T	Moderate	Measure once
Aortic valve	Aortic stenosis score	p89-95	Moderate	Baseline echocardiogram
Hypertrophic cardiomyopathy	Polygenic score	p76-89	Moderate	Baseline echocardiogram
Venous thrombosis	Score; loci; F11	p86.3; p95.1; F11 T/T	Moderate	Prophylaxis in surgery, immobilization and long journeys
Eye	Glaucoma	p97.9	Moderate-high	OCT + intraocular pressure
Lung	Emphysema/COPD	p96.2 (+ Pi*MZ)	Moderate	Base spirometry; zero smoking
Hearing	Tinnitus	p99.8	Moderate	Base audiogram
Colorectal	Cancer, polyps, early onset	p94.3; p92.7; p82.3	Moderate	FIT from 40; colonoscopy to 45
Bladder	Bladder cancer	p94.9 (+ NAT2 slow)	Moderate	Not using pioglitazone; hematuria investigated
Bone and muscle	Fractures; fragility; lean mass	p84; p75; p10	Moderate	DEXA base; strength training
Mental health and sleep	Depression; ADHD; insomnia	p95; p94/p97; p89	Moderate	Mood tracking (also with isotretinoin); TCC for insomnia
Immunity	Sarcoidosis; vitiligo; uveitis	p92.3; p89.9; p84.5	Low moderate	Signal surveillance only
Melanoma	Chip score	p6.5 / p25	Moderate	Photoprotection (isotretinoin); no extra screening by genetics
Autoimmunity	HLA clean (without B*27, C*06:02, DQ2.5, DQ8, DRB1*15:01, DRB1*04)	Chip tags	High (tags)	Low risk of celiac, spondylitis, psoriasis and EM

3. Favourable

- **No pathogenic variants** in genes of cardiomyopathy, arrhythmia, hereditary cancer and metabolic disease that the chip reads, in addition to Pi*Z.
- **Low global coronary score** (p5) despite 9p21.
- **Clean autoimmune HLA**; PTPN22 and STAT4 without risk; IBD p1,7, celiac p3,9, psoriasis p2.4-6.3.
- **Negative classical hereditary thrombophilia** (F5 Leiden, prothrombin G20210A).
- **APOE ε3/ε3**; Homozygous longevity FOXO3; longevity score p92.
- **DPYD, TPMT, NUDT15, G6PD, CYP2C19, BCHE**; RYR1 and CACNA1S without detected variants.
- **Hepatitis A**: already immune (do not need a vaccine).

4. Unfavourable

- **Probable Pi*MZ** + PNPLA3 + MBOAT7 (liver and lung).
- **Homozygous 9p21** and CETP of high activity.
- **High scores:** glaucoma, emphysema, tinnitus, colorectal, bladder, aortic stenosis, venous thrombosis, fractures, depression, ADHD, insomnia.
- **Pharmacogenetics:** CYP2C9 *1/*3 + VKORC1 T/T (very high sensitivity to coumarins), NAT2 slow UGT1A1 *28 (Gilbert).
- **Probable null GSTM1** and GSTA1 *B/*B (weaker II phase detoxification).

5. Pharmacogenetics (summer)

Gene	Genotype	Consequence (suggestion, not applied)
CYP2C9	*1/*3	Avoid piroxicam/tenoxicam; meloxicam 50%; NSAID in minimum dose and short course; phenytoin –25%; siponimod 1 mg; fluvastatin ≤ 40 mg
VKORC1 + CYP2C9	–1639 T/T + *3	Prefer direct anticoagulant; coumarins only with a very low genetic dose
NAT2	Slow *5/*5	Isoniazid with hepatic surveillance; hydrolazine 40–75 mg/day; sulfonamides, dapsone, procainamide with surveillance
UGT1A1	*1/*28	Irinotecan/sacituzumab: neutropenia; moderate dose paracetamol
CYP3A5	*1/*3 expressor	Tacrolimus 1.5–2; sirolimus by 24–48 h dosage after a dose
SLCO1B1	Probable *1/*14 (augmented function)	All statins at the desired dose
CYP2D6	Probable *1/*2, without copy number	Confirm before codeine, tramadol, tamoxifen, eligostat, pimozide
HLA-B*58:01	Probable negative (imputed)	Tip before allopurinol
Pioglitazone	—	Not recommended for genetic reasons (beak p94.9, slow NAT2, reduced ejection fraction score p92)
Anesthesia	RYR1/CACNA1S without variants; BCHE normal	Standard precautions; thrombosis prophylaxis (p86.3)

6. Analysis and exams to be asked

All lines of this section are suggestion (not applied); the decision to request each exam is from the doctor.

Now

Examination	Genetic motif
Serum Alpha-1-antitrypsin + Pi phenotype	Confirm Pi*MZ (rs28929474 C/T)
Hepatic elastography (baseline, then periodic)	Pi*MZ + PNPLA3 I148M + MBOAT7
Lp(a) , once in life	High genetics unlikely, but not excluded by the chip
HLA typing of 2 fields	HLA-B*58:01 (allopurinol), A*31:01, B*15:02; confirm the inferred HLA
Echocardiogram	Aortic valve (p89-95 aortic stenosis) and genetics of hypertrophic cardiomyopathy (p76-89)
OCT of the optic nerve + intraocular pressure	Glaucoma p97.9
Spirometry	P96.2 + Pi*MZ
Audiogram	Tinnitus p99.8
DEXA (bone density and body composition)	P84 fractures, p75 fragility, p10 lean mass
Serolimus 24-48 h dosage after a dose	CYP3A5 expresser (less sirolimus per mg)

Vaccines

Vaccine	Reason
Hepatitis B	Hepatic protection (Pi*MZ + PNPLA3)
Conjugated pneumococcal	Rapamycin immunomodulation; high pneumonia score
Influenza (annual) and COVID-19	Immunomodulation; MBL2 and IFIH1
Recombinant zoster	Rapamycin immunomodulation
Hepatitis A	It is not necessary: already immune

Next years

Examination	When	Reason
Investigation of occult blood in feces (FIT)	Annual from 40	Colorectal p94.3; p92.7 polyps
Colonoscopy	Up to 45, at most	Colorectal and early onset p82.3
Genome complete to 30x (clinical)	Next years	CYP2D6, HLA, GSTM1/GSTT1, DNA mitochondrial, ACMG genes and structural variants that the chip does not read
Repeat elastography, OCT and audiogram	As the base result	—

Later

Examination	Reason
Coronary calcium score	Homozygous 9p21 (5 repeat in 5 years if 0)
Clonal hematopoiesis panel (CHIP)	Somatic finding that the chip does not read; more useful with age
General age screenings (abdominal aorta to 65 in men, among others)	General recommendations, without proper genetic justification

Also to consider

- **Research of H. pylori**, once (PSCA T/T, IL1B -511T, CXCL8 -251 A/A). *Suggestion (not applied).*
- **Basal serum tryptase**, once, only if there are symptoms of mast cells (proxy of alpha-tryptasemia absent). *Suggestion (not applied).*
- **Homocysteine with erythrocyte folate and methylmalonic acid** (MTHFR compound heterozygous; p92 homocysteine score). *Suggestion (not applied).*

7. Findings for confirmation and limits of the chip

- **Six homozygous (or hemizygous) calls of ultra-rare variants** in GBA1, IDUA, ALG1, NF1, GOSR2 and OFD1: almost certainly chip artifacts (without compatible phenotype and absent from the reference panel). They should not motivate decisions; the complete genome clarifies them.
- **The chip reads about 7% of known pathogenic variants:** the absence of findings does not exclude hereditary disease.
- **Unread:** CYP2D6 copy number, GSTM1/GSTT1 deletions, HLA to 2 fields, KIV-2 repeat of Lp(a), DNA complete mitochondrial (including MT-RNR1), structural variants, hereditary alpha-tryptasemia.
- **Polygenic scores:** Relative risk, with large overlap among people; were calculated in the reference population and may be less accurate outside of it.

8. Map for aging mechanisms

Mechanism	Genetic finding	Meaning
Vascular Aging	Homozygous 9p21, high CETP; coronary score p5	mixed
Remodelling of the extracellular matrix	Pi*MZ (elastase); p89-95 aortic stenosis	unfavourable
Fragmentation of Elastin	Probable Pi*MZ	unfavourable
Metabolic Homeostasis Decline	PNPLA3, MBOAT7 (fat liver); HSD17B13 protector	unfavourable
Chronic Inflammation	Calm-based cytokines; CARD9 and reactive gastric pathway	mixed
Somatic Mutation Accumulation	Colorectal p94.3; bladder p94.9	unfavourable
Clonal hematopoiesis (CHIP)	Not read in the chip; panel later	without reading
Osteoporosis	P84 fractures	unfavourable
Thin Mass Loss	P10 lean mass; p75 fragility	unfavourable
Sleep disturbance	Insomnia p89	unfavourable
Neuroinflammation	APOE ε3/ε3; p95 depression	mixed
Redox imbalance	GSTM1 null, GSTA1 *B/*B; SOD2, NQO1, NRF2 favourable	mixed

9. Limitations and methods

- **Methods:** chip genotypes (strand +, GRCh38) and local imputation with a public panel of genomes complete to 30x ($DR2 \geq 0.8$); PGS Catalog scores with percentiles in the reference population; pharmacogenetic according to CPIC/DPWG guidelines and FDA/EMA labels. No genetic data came out of the analysis computer.
- The findings come from the thematic reports of Sérgio F (pharmacogenomic, anesthesia, cardiovascular, liver, cancer, immunity, among others) and were reconfirmed against raw DNA and imputation.
- It is a genetic summary: it does not replace the clinical history, physical examination or analysis.
- Conclusions only from genes. All actions: suggestion (not applied).